

Release_Checklist_Standard

Technical Standard

保险公司 技术开发部
内部文档 · 仅供授权人员查阅

Contents

- 1. Overview
 - 1.1 Purpose
 - 1.2 Scope
- 2. Release Types & Process
 - 2.1 Release Types
 - 2.2 Release Flow Overview
- 3. Pre-Release Checklist
 - 3.1 Code Review
 - 3.2 Test Verification
 - 3.3 Environment Preparation
 - 3.4 Rollback Plan
 - 3.5 Canary Strategy
 - 3.6 Notification & On-Call
- 4. Canary Release Standards
 - 4.1 Canary Strategy Template

Canary Release Plan - [Project] v[Version]

 - Canary Scope
 - Verification Metrics
 - Promote Conditions
 - Rollback Conditions
 - 4.2 Canary to Full Rollout
- 5. During-Release Checklist
 - 5.1 Database Changes
 - 5.2 Application Deployment
 - 5.3 Release Confirmation
- 6. Post-Release Checklist
 - 6.1 Observation Period (30 min - 24 hours)
 - 6.2 Confirmation & Retrospective
- 7. Appendix
 - 7.1 Reference Documents

Release Checklist Standard

> Reference: Tencent Release Standards, Alibaba Cloud Release Process, Google SRE Change Management

1. Overview

1.1 Purpose

Standardize pre-release checklists to ensure all preparations are in place, risks are controlled, and rollback plans are complete, reducing the probability of production incidents caused by releases.

1.2 Scope

Applies to all production environment releases within the Technology Development Department, including feature releases, hotfixes, configuration changes, and data migrations.

2. Release Types & Process

2.1 Release Types

Type	Definition	Approval Chain	Window
Regular	Feature iteration, bug fixes	Tech Lead + QA	Weekly fixed window
Emergency	P0/P1 defect fix	Tech Lead + Architect	Emergency window (24h)
Configuration	Config center change	Config admin	Business hours
Data	Data correction, migration	DBA + Tech Lead	Change window

2.2 Release Flow Overview

Code Freeze > Test Report > Release Approval > Environment Prep > DB Change > App Deploy > Canary > Full Release > Observation

3. Pre-Release Checklist

3.1 Code Review

- ☐ Code reviewed and approved
- ☐ All unit tests passed (coverage meets standard)
- ☐ Static analysis: no P0/P1 issues (SonarQube / CodeQL)
- ☐ Security scan: no high-risk vulnerabilities (SAST / DAST)
- ☐ Dependency check: no known vulnerabilities (SCA / Dependabot)
- ☐ Code merged to target branch (release / master)
- ☐ Git tag created (format: v{major}.{minor}.{patch})

3.2 Test Verification

- ☐ Functional test report signed off

- ☐ Integration tests passed
- ☐ Regression tests passed (core cases 100%)
- ☐ Performance test passed (or assessed as non-impacting)
- ☐ UAT accepted (business confirmation)
- ☐ Compatibility tests passed (browser/API version/DB version)

3.3 Environment Preparation

- ☐ Staging environment verified
- ☐ Production configuration reviewed (config diff)
- ☐ Database change scripts approved and pre-verified
- ☐ External service versions confirmed (third-party APIs/middleware)
- ☐ SSL certificate valid
- ☐ CDN cache refresh plan confirmed
- ☐ DNS changes completed

3.4 Rollback Plan

- ☐ Rollback plan documented
- ☐ Application rollback steps verified (rollback script to previous version)
- ☐ Database rollback script prepared and verified
- ☐ Rollback trigger conditions defined
- ☐ Estimated rollback time assessed (RTO)
- ☐ Post-rollback data validity verified

3.5 Canary Strategy

- ☐ Canary release plan finalized
- ☐ Canary ratio and timing confirmed
- ☐ Canary verification metrics defined (error rate, RT, business metrics)
- ☐ Canary rollback conditions defined
- ☐ Monitoring configured during canary

3.6 Notification & On-Call

- ☐ Release notification sent
- ☐ Affected system owners notified
- ☐ On-call personnel confirmed (online during release)
- ☐ Customer support/operations team informed of potential impact
- ☐ Business stakeholders aware of release window

4. Canary Release Standards

4.1 Canary Strategy Template

```
## Canary Release Plan - [Project] v[Version]
### Canary Scope
- Ratio: [ ]% (recommend starting at 1%)
- Target: by [channel / region / user ID hash]
- Duration: [ ] hours/days
### Verification Metrics
- Error rate: P0 API error rate < 0.1%
- P99 RT: < 1.2x baseline
- Business metrics: no degradation in policy/payment success rates
- System resources: CPU < 70%, memory < 80%
### Promote Conditions
```

1. No P0/P1 alerts after [] hours of canary
 2. Core business metrics normal
 3. Performance metrics within threshold
- ### Rollback Conditions
1. Error rate exceeds []%
 2. P0 API unavailable
 3. Key business metrics significantly degraded (e.g., 10%)
 4. Financial loss detected

4.2 Canary to Full Rollout

Step	Action	Verification	Duration
1	Canary 1%	Metrics normal	30 min
2	Canary 5%	Metrics normal	30 min
3	Canary 20%	Metrics normal	1 hour
4	Canary 50%	Metrics normal	1 hour
5	Full 100%	Metrics normal	30 min observe

5. During-Release Checklist

5.1 Database Changes

- ☐ Backup affected tables
- ☐ Pre-verify rollback script on a replica
- ☐ Execute DDL in order (least dependencies first)
- ☐ Verify after each step (SHOW TABLES, DESC)
- ☐ Pre-count DML data changes

5.2 Application Deployment

- ☐ Drain old version traffic
- ☐ Deploy new version (rolling update / blue-green)
- ☐ Health check passes after startup (/health)
- ☐ Restore traffic for observation
- ☐ Confirm metrics on monitoring dashboard

5.3 Release Confirmation

- ☐ Release system status: OK
- ☐ Core API smoke test passed
- ☐ No abnormal alerts
- ☐ No ERROR level logs
- ☐ Business confirms functionality

6. Post-Release Checklist

6.1 Observation Period (30 min - 24 hours)

- ☐ Monitor dashboard continuously
- ☐ Compare business metrics (pre vs post release)
- ☐ No abnormal error log growth
- ☐ No abnormal slow queries
- ☐ No capacity issues

6.2 Confirmation & Retrospective

- ☐ Release result notification (success/failure/rollback)
 - ☐ Update change record
 - ☐ Archive release artifacts (package/image tag)
 - ☐ Initiate postmortem if issues occurred
-

7. Appendix

7.1 Reference Documents

- *Release & Change Management Standard*
- *Database Change Management Standard*
- *Emergency Response Plan & SOP*
- Environment configuration checklists

7.2 Rollback Plan Template

```
# Rollback Plan – [Project] v[Version]
## Trigger Conditions
1. [Condition 1]
2. [Condition 2]
## Application Rollback
1. Rollback to previous version image: [Image Tag]
2. Deploy command: [Command]
3. Verification: [Health check command]
## Database Rollback
1. Rollback script: [SQL file path]
2. Execution method: [DBA / automation tool]
3. Verification: [Verification SQL]
## Data Compensation
– If data was written, compensation plan: [Description]
## Post-Rollback
– Notify stakeholders
– Document rollback reason
– Improvement plan
```